YACHAY TECH UNIVERSITY

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Solving Recurrences: Linear Second Order

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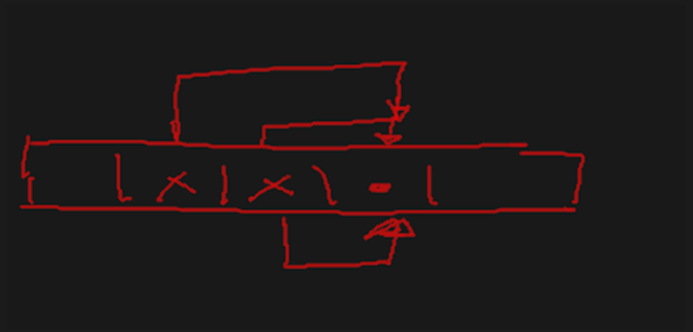
Linear Second Order Recurrences

General form:

$$a\,x(n) + b\,x(n-1) + c\,x(n-2) = f(n)$$

second-order linear recurrence with constant coefficients



if f(n) = 0 ----> homogeneous
if f(n) != 0 ----> inhomogeneous

Solving Second Order Recurrences

Method:

1. Find the characteristic equation
2. Find the characteristic roots
3. Set up the framework

$$\mathrm{x(n)} = C_1 r_1^n + C_2 r_2^n$$

4. Use I.C. to find C_1 and C_2

$x(n) = 3\,x(n-1) - 2\,x(n-2)$

$x(0) = 0$

$x(1) = 1$

if $r_1 \mathrel{!=} r_2$

$x(n) = c_1 r^n + c_2 r^n$

Case 1

Try: $x(n) = r^n$

$$\frac{r^n}{r^{n-2}} = \frac{3 \cdot r^{n-1}}{r^{n-2}} - 2\,\frac{\cancel{r^{n-2}}}{\cancel{r^{n-2}}}$$

$$r^n \cdot r^{-(n-2)} = 3 \cdot r^{n-1} \cdot r^{-(n-2)} - 2$$

$$r^{n-n+2} = 3\,r^{n-1-n+2} - 2$$

$$r^2 = 3r^1 - 2 \Rightarrow \boxed{r^2 - 3r + 2 = 0}$$

C.E.

$$r^2 - 3r + 2 = 0$$

$$(r-2)(r-1) = 0$$

$$\boxed{r_1 = 2} \quad \boxed{r_2 = 1}$$

C.R.

$$\Rightarrow x(n) = c_1 r_1^n + c_2 r_2^n$$

$$x(0) = 0 \qquad x(1) = 1$$

$$x(0) = c_1 r_1^0 + c_2 r_2^0 \qquad x(1) = c_1 r_1^1 + c_2 r_2^1$$

$$x(0) = c_1 + c_2 \qquad x(1) = c_1 r_1 + c_2 r_2$$

$$c_1 + c_2 = 0 \qquad c_1 r_1 + c_2 r_2 = 1$$

$$\begin{cases} c_1 + c_2 = 0 \\ 2c_1 + c_2 = 1 \end{cases} \Rightarrow \boxed{c_2 = -1} \quad \boxed{c_1 = 1}$$

$$x(n) = 1 \cdot 2^n + (-1) \cdot 1^n$$

$$\boxed{x(n) = 2^n - 1}$$

$x(n) = 5\,x(n-1) - 6\,x(n-2)$

$x(0) = 1$

$x(1) = 0$

$x(n) = r^n$

$$\frac{r^n}{r^{n-2}} = \frac{5r^{n-1}}{r^{n-2}} - \frac{6\cancel{r^{n-2}}}{\cancel{r^{n-2}}}$$

$\boxed{r^2 - 5r + 6 = 0}$ C.E.

$(r-3)(r-2) = 0$

$\boxed{r_1 = 3}$ $\boxed{r_2 = 2}$ C.R.

$x(n) = c_1 r_1^n + c_2 r_2^n$

$x(0) = 1$

$x(0) = c_1 r_1^0 + c_2 r_2^0$

$x(0) = c_1 + c_2$

$c_1 + c_2 = 1$

$x(1) = 0$

$x(1) = c_1 r_1^1 + c_2 r_2^1$

$x(1) = 3c_1 + 2c_2$

$3c_1 + 2c_2 = 0$

$$\begin{cases} c_1 + c_2 = 1 \\ 3c_1 + 2c_2 = 0 \end{cases}$$

$c_1 = 1 - c_2$

$3(1 - c_2) + 2c_2 = 0$

$c_1 = 3$ $\quad c_2 = -2$

$x(n) = 3 \cdot 2^n - 2 \cdot 3^n$ //

Fibonacci Numbers

F(n) - F(n-1) - F(n-2) = 0

$$F(n) = F(n-1) + F(n-2)$$

$$\frac{r^n}{r^{n-2}} - \frac{r^{n-1}}{r^{n-2}} - \cancel{\frac{r^{n-2}}{r^{n-2}}} = 0$$

$$r^2 - r - 1 = 0 \quad \text{C.E.}$$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 1 \quad b = -1 \quad c = -1$$

$$r = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1}$$

$$r = \frac{1 \pm \sqrt{5}}{2} \quad \text{C.R}$$

$$x(n) = c_1 r_1^n + c_2 r_2^n$$

$$= c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$\text{I.C.} \quad F(0) = 0 \qquad F(1) = 1$$

$$c_1 = \frac{1}{\sqrt{5}} \qquad c_2 = -\frac{1}{\sqrt{5}}$$

$$x(n) = \frac{1}{\sqrt{5}} \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \cdot \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$x(n) = \frac{1}{\sqrt{5}} \phi^n - \frac{1}{\sqrt{5}} \phi^n \quad //$$

Case Two Linear Second Order Recurrences

if $r_1 = r_2$

$x(n) = c_1 r^n + c_2 n r^n$

$x(n) - 6\,x(n-1) + 9\,x(n-2) = 0$

$x(0) = 0$

$x(1) = 3$

$r^n - 6r^{n-1} - 9r^{n-2} = 0$

$r^2 - 6r + 9 = 0$ C.E

$(r-3)(r-3) = 0$ C.R.

$r_1 = 3 \quad r_2 = 3$

$x(n) = c_1 r^n + c_2 n r^n$

$x(0) = c_1 \cdot 3^0 + c_2 \cdot 0 \cdot 3^0$

$x(0) = c_1$

$x(1) = c_1 \cdot 3^1 + c_2 \cdot 1 \cdot 3^1$

$= 3c_1 + 3c_2$

$c_2 = 1$

$x(n) = 0 \cdot 3^n + 1 \cdot n \cdot 3^n$

$x(n) = n \cdot 3^n$ //

Recurrences: Examples

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Example 1:

$x(n) = x(n-1) + 5$ for all $n>1$

$x(1) = 0$

Forward Substitution $\longrightarrow$ Backward Substitution

$x(1) = 0 \quad = 0$

$x(2) = 0 + 5 = 5$

$x(3) = 5 + 5 = 10$

$x(4) = 10 + 5 = 15$

$\vdots$

$\downarrow$

?

Guess!

$x(n) = (5 \cdot n) - 5$

prove! M.I.

$x(n) = 5(n-1)$

prove! M.I.

Example 2:

x(n) - x(n-1) - 6x(n-2) = 0

x(0) = 1

x(1) = 8

$$x(n) = c_1 r_1^n + c_2 r_2^n$$
$$= c_1 3^n - c_2 2^n$$

$x(0) = 1$

$x(0) = c_1 3^0 - c_2 2^0$

$c_1 - c_2 = 1$

$x(1) = 8$

$x(1) = c_1 3^1 - c_2 2^1$

$3c_1 - 2c_2 = 8$

$$\frac{r^n}{r^{n-2}} - \frac{r^{n-1}}{r^{n-2}} - \frac{6\cancel{r^{n-2}}}{\cancel{r^{n-2}}} = 0$$

$\boxed{r^2 - r^1 - 6 = 0}$ C.E.

$(r-3)(r+2) = 0$

$\boxed{r_1 = 3}$ C.R

$\boxed{r_2 = -2}$

$r_1 \neq r_2$

$$\begin{cases} c_1 - c_2 = 1 \\ 3c_1 - 2c_2 = 8 \end{cases}$$

$c_1 = 1 + c_2$

$3(1 + c_2) - 2c_2 = 8$

$3 + 3c_2 - 2c_2 = 8$

$\boxed{c_2 = 5}$

$c_1 = 1 + c_2$

$\boxed{c_1 = 6}$

$$\boxed{x(n) = 6 \cdot 3^n - 5 \cdot 2^n}$$

Q & A

FOR QUESTIONS OR COMMENTS

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